Restoring the raw memory descendants
A two-chaos residual for the second memory coefficient

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Local research note — 6 November 2026

Abstract

We restore the lower Gaussian-chaos components of the four previously retained quartic descendants. Their second-chaos limits contain no output-deficit direction. A fixed-charge trace argument transfers the earlier fourth-chaos Gram to half filling. Consequently the raw second memory coefficient, after projection onto a specified finite span, retains both the known fourth-chaos residual and the newly resolved second-chaos residual. This completes their equal-time assembly without assuming that their energy-resolved cross vanishes.

1 State, scale, and result

Use standard CAR and the full overlapping complex Gaussian p-body Hamiltonian, $\hbar = 1$, fixed charge $m = N/2$, normalized maximally mixed trace, and $\langle X, Y\rangle = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger Y$. All projections use this annealed inner product. Write $h = H_m$, $k = H_{m-1}$ with the same couplings, $R_j = \mathrm{Rem}(k^j)$, and

$$A = R_2 + (2p - m)h^2, \quad B = \mathsf{W}_2 h^2, \quad O = \mathsf{W}_2 A, \qquad J_h X = \{h, X\}/2. \tag{1}$$

Take $\lambda_N = p^2/N \to \lambda > 0$, $q_N = e^{-4\lambda_N}$, $C_N = \mathbb{E}\,\mathrm{tr}_m\, h^2 \to \sigma^2 = J_s^2/\lambda$. Lower-chaos coefficients retain C_N, λ_N, q_N until after their growing terms are subtracted. Let

$$\mathcal{V}_N = \mathrm{span}\{I, B, O, J_h F, E_0, M_0, S_0, Z_0\}, \qquad \widehat{\Pi}_N = \mathrm{proj}_{\mathcal{V}_N}, \tag{2}$$

where the four raw quartic descendants are defined on the next page. With $\Pi_N = \mathrm{proj}_{\{I,B,O,J_hF\}}$ and $T_N = (1 - \Pi_N)\mathcal{S}_N$, the result is

$$(1 - \widehat{\Pi}_N) m Z_2 = Q_{4,N} + 8\lambda_N C_N q_N^2 T_N + o_{L^2}(1). \tag{3}$$

$Q_{4,N}$ is the preceding fourth-chaos residual and $\mathcal{S}_N$ is the second-chaos output-deficit insertion. Their equal-time inner product is exactly zero by Gaussian-chaos orthogonality. Thus the limiting norm squared is $V_4 + V_2 > 0$, with explicit expressions below. The claim concerns this operator span, not the full algebra of conserved quantities.

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2 Five raw words and exact low-chaos restoration

Define $\mathcal{T}_j = R_j + (jp - m)h^j$. The five raw operators are

$$\begin{aligned} E_0 &= \{A, h^2\}/2, & M_0 &= hAh, \\ S_0 &= \{h, \mathcal{T}_3\}/2 - E_0 - M_0, & C_0 &= \mathcal{T}_4 - 2E_0 - M_0 - 2S_0, \\ Z_0 &= h^4. \end{aligned} \tag{4}$$

They are linear combinations of $\mathbf{R} = (R_4, \{h, R_3\}, \{h^2, R_2\}, hR_2h, h^4)^T$:

$$\begin{pmatrix} E_0 \\ M_0 \\ S_0 \\ C_0 \\ Z_0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1/2 & 0 & 2p-m \\ 0 & 0 & 0 & 1 & 2p-m \\ 0 & 1/2 & -1/2 & -1 & m-p \\ 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \mathbf{R}. \tag{5}$$

In particular, the memory core satisfies the exact identity

$$\mathcal{E} = C_0 - 2S_0 + (E_0 + M_0)/2. \tag{6}$$

The single-contraction map for a homogeneous quartic polynomial is $\mathfrak{C} = \frac{1}{2}\sum_i v_i \partial_{x_i}^2$, where x_i are independent real coupling coordinates of variance v_i. Hence

$$e_P = \mathbb{E}P, \qquad \mathsf{W}_2 P = \mathfrak{C}P - 2e_P, \qquad \mathsf{W}_4 P = P - \mathfrak{C}P + e_P. \tag{7}$$

Apply (5) to the preceding five exact contractions $(D_4, D_{13}, D_{22}, D_{121}, D_{40})^T$ to obtain every $\mathfrak{C}P$. This uses the same-coupling adjacent-charge covariance, not independent Hamiltonians [1, 3].

For clarity, each scalar mean can also be recovered from the rank filters. If $\mathfrak{C}P = \sum(g_P^+ Z^+ + g_P^- Z^-)$, then in the common-rank domain $p \le \min(m-1, N-m)$,

$$e_P = \frac{1}{2}\sum_{a=0}^{p} \left[C_a g_P^+(a,a,0) + m C_a^- g_P^-(a,a,0) \right] I. \tag{8}$$

Here $C_a = \mathbb{E}\,\mathrm{tr}_m\, h_a^2$ and $C_a^- = \mathbb{E}\,\mathrm{tr}_{m-1}\, k_a^2$. The common-rank condition holds eventually in the present limit. Equations (5)–(8) restore $\mathsf{W}_0, \mathsf{W}_2, \mathsf{W}_4$ at finite N; scalar subtraction is not left implicit.

3 The complete second-chaos descendant table

For $P = E_0, M_0, S_0, C_0, Z_0$, write

$$\mathsf{W}_2 P = C_N[p\ell_P + k_P]B + C_N o_P O + j_P J_h F + C_N s_P \mathcal{S}_N + o_{L^2}(1). \tag{9}$$

The table uses $\lambda = \lambda_N$ and $q = q_N$ on its right-hand side, so the coefficients of order p retain finite-N parameters.

P	ℓ_P	k_P	o_P	j_P	s_P
E_0	2	$-2\lambda q^2$	1	0	0
M_0	2	$-2\lambda(2q-1)$	q^2	0	0
S_0	$2q$	$-2\lambda(q^2-q+1)$	q	1	0
C_0	$2q^2$	$-2\lambda(8\lambda q^2 - 3q^2 + 2q + 2)$	1	$2q$	$8\lambda q^2$
Z_0	0	$q^2 + 2q + 3$	0	0	0

This is obtained by applying (5) to the exact filters, then changing from (Z^+, Z^-) to $(Z^+, A_{ab;r})$ before expanding. The full rational coefficients and both fixed-deficit and rank-proportion expansions are saved in the symbolic certificate.

Here is the limit justification for all five rows. Each A-filter has bounded rational prefactors multiplying $1, \kappa_a/C_N, \kappa_b/C_N, \kappa_r/C_N$. Their adjacent-rank difference is $O(p^{-1})$, giving

$$|g_{A,P}/C_N - o_P| \le K_\lambda(1 + u + v + S)/p, \quad |g_{A,P}/C_N| \le K_\lambda. \tag{10}$$

The preceding input-weighted A norm and sixth ordinary-product moment bound therefore apply unchanged: a window $S \le p^{3/4}$ gives an $O_{L^2}(p^{-1/4})$ error for this part.

Each H-filter, after subtracting its leading $p\ell_P$, has a uniform bound $K_\lambda(1 + u + v + S)$ and a fixed-deficit affine limit. The previous factorial input tail and third output moment give uniform integrability of its square. Thus the H-part converges in L^2. Finally, $\mathsf{W}_2\{h, D_N\} = L_{\mathrm{in}} + 2\bar{s}_N B$ and $J_h F - 4\lambda_N q_N C_N L_{\mathrm{in}} \to 0$ give the centered coefficients k_P, j_P. The total remainder in (9) is only claimed to be $o_{L^2}(1)$.

The decisive entry is the last column. Every old raw descendant has $s_P = 0$; restoring them cannot absorb the output-deficit direction through an additional second-chaos component.

4 Transferring Wick-block moments to half filling

The earlier marked-word calculation used $2^{-N}\,\mathrm{Tr}$. A bridge is required before using its fourth-chaos Gram here. For a diagonal occupation constraint on L distinct sites, with t occupied and $L-t$ empty, the ratio of fixed half-filled trace to full-Fock trace is exactly

$$\mathcal{R}_N(L,t) = \frac{2^L (m)_t^{\downarrow} (m)_{L-t}^{\downarrow}}{(N)_L^{\downarrow}} = 2^L \frac{\binom{N-L}{m-t}}{\binom{N}{m}}. \tag{11}$$

Uniformly for $L \le kp = O(\sqrt{N})$ and allowed t,

$$\log \mathcal{R}_N(L,t) = \frac{L-(2t-L)^2}{2N} + O(L^3/N^2), \qquad \mathcal{R}_N(L,t) \le \exp(K_k p^2/N). \tag{12}$$

The expansion follows from the three falling factorials; the upper bound also follows directly by dropping their numerator factors after extracting powers of m. It is valid for large N at fixed word length k.

In a Gaussian-paired word, every chord carries two independent p-element index sets. At a site belonging to only one such set, the paired creation/annihilation entries force its initial occupation. The two sets of each chord have opposite singleton occupations and equal cardinality. If K is the total pairwise intersection count, deleting the multiply used sites changes their balance by at most a constant times K. Therefore $|2t-L| \le c_k K$ for every nonzero local occupation term. The unmarked word has $L \le kp$.

For the marked Wick blocks used here, no Gaussian chord is internal to a block. Each body-rank mark hence contracts indices from different chords. It is supported on their intersections and leaves the singleton argument intact. Expanding the multiply used sites in occupation projectors has absolute coefficients bounded by $e^{c_k K}$ times a fixed polynomial in K. This is also the absolute overlap bound used in the preceding marked-word derivation.

First restrict $K \le K_0$. Equations (11)–(12) give $\mathcal{R}_N \to 1$ uniformly. The established exponential moment bounds for intersections of a fixed number of random p-element sets control the complementary absolute tail, including the marks; the global bound in (12) preserves it. Letting $K_0 \to \infty$ proves equality of the two trace limits.

This transfers the fourth-chaos Gram and its fixed energy moments. It also transfers the bounded fixed-degree marked Wick-block moments needed for the centered fields below. It does not identify ordinary coupling chaos with the spectral q-Wick polynomials of [2]; those are different constructions.

5 Projection after restoring both chaos sectors

Let $D_i = (1 - \Pi_N)P_i$ for $P_i = E_0, M_0, S_0, Z_0$. Their means are scalar and their second-chaos parts lie in $\mathrm{span}\{B, O, J_h F\}$ up to $o_{L^2}(1)$. Thus

$$D_i = \mathsf{W}_4 P_i + o_{L^2}(1). \tag{13}$$

Their limiting Gram is the previously computed fourth-chaos nuisance Gram, now justified in the same half-filled trace. It is strictly positive. With Gram entries divided by σ^8, its determinant is

$$\begin{aligned} &16\lambda^3(1-q)^2(1+q)^4(1+q^2)d_3 d_5(1+q+4\lambda q)^2 \\ &\times [1 - q^3 + 4\lambda q^2(2q^2 - q + 2)] > 0, \quad d_3 = q^2 - q + 1,\ d_5 = 1 + q + q^2 + q^3 + q^4. \end{aligned} \tag{14}$$

The fourth-chaos residual is

$$\begin{aligned} Q_{4,N} &= \mathsf{W}_4(C_0 - \alpha E_0 - \beta M_0 - \gamma S_0 - \delta Z_0), \\ \alpha &= -\frac{2q^3}{d_3 d_5}, \quad \beta = -\frac{q^2(1+q+q^2)}{d_3 d_5}, \quad \gamma = \frac{2q}{d_3}, \\ \delta &= \frac{2\lambda(1-q)^3(1+q)(1+q+q^2)}{(1+q^2)d_5}. \end{aligned} \tag{15}$$

These coefficients can be evaluated at λ_N, q_N. After applying $1 - \Pi_N$, the core is bounded and equals its fourth-chaos component plus $8\lambda_N C_N q_N^2 T_N + o_{L^2}(1)$. The latter component is exactly orthogonal to every $\mathsf{W}_4 P_i$. Invertibility of the Gram and continuity of this finite-dimensional projection prove (3) for $\mathcal{E}$ in place of mZ_2.

The two limiting contributions are

$$V_4 = \frac{J_s^8}{\lambda^4} \frac{4\lambda(1-q)^2(1+q)^2[P(q) + 4\lambda q^3 U(q)]}{(1+q^2)d_3 d_5}, \tag{16}$$

$$P(q) = 1 + q + 2q^2 + q^3 + q^4, \quad U(q) = 2 + q + 2q^2 + q^3 + 2q^4,$$

$$V_2 = \frac{64 J_s^8 q^4(1+q)(1-q^2+8\lambda q)}{\lambda(1+q+4\lambda q)}. \tag{17}$$

Both are positive for fixed $\lambda > 0$. After the double-scaled limit, $V_4 \sim 768 J_s^8/(5\lambda)$ and $V_2 \to 1024 J_s^8$. These are energy-to-the-eighth variances of the collective coefficient mZ_2, not of unscaled Z_2.

6 From the core to the raw second memory coefficient

For completeness, the second centered memory coefficient is defined from $\widehat{M}(s) = G + isZ_1 - s^2 Z_2/2 + O(s^3)$ at finite N. The exact first-return calculation gives, with $r = p/m$,

$$mZ_2 = B_{22}^f + \frac{(1+r)^2}{4}[h,[h,A]] + 2rJ_hT_3 + 3r^2J_h^2A - A^2/m - mr^4h^4. \tag{18}$$

Here T_3 is the Hermitian cubic derivative singlet and $B_{22}^f = \sum_i (\mathcal{L}^2 c_i)^\dagger \mathcal{L}^2 c_i$. Use the previous centered fields

$$K_A = a_m + c_m(h^2 - C_N), \quad Y = A - K_A, \quad P_m = t_{3,m}h^3 + t_{1,m}h, \quad U = T_3 - P_m. \tag{19}$$

c_m is the exact energy-quadratic regression coefficient; P_m is the preceding cubic conserved subtraction. Its precise scalar coefficients do not enter the projection below. Define

$$D_{2,m} = 2rJ_hP_m + 3r^2J_h^2K_A - K_A^2/m - mr^4h^4. \tag{20}$$

It belongs exactly to $\mathrm{span}\{I, h^2, h^4\} \subset \mathcal{V}_N$. The remaining identity is

$$\begin{aligned} mZ_2 - D_{2,m} - \mathcal{E} = (r/2 + r^2/4)[h,[h,Y]] + 2rJ_hU + 3r^2J_h^2Y \\ - (\{K_A, Y\} + Y^2)/m. \end{aligned} \tag{21}$$

The centered Wick-block representations of Y, U transfer by the argument on page 4. Their linear filtered pieces are also bounded: for an invariant Hermitian Gaussian field X with $\mathbb{E}X^2 = vI$, Wick pairing and Cauchy–Schwarz give $\mathbb{E}\,\mathrm{tr}\,X^4 \le 3v^2$; their v is controlled by the previous input-rank tails. The cubic Wick blocks and the products with Y have the preceding absolute marked-word bounds. Thus the fixed-sector moments needed for h^2Y, hU, Y^2 are bounded. Noncommutative Hölder uses, for example, $\mathbb{E}\,\mathrm{tr}\,h^8$, $\mathbb{E}\,\mathrm{tr}\,Y^4$, and $\mathbb{E}\,\mathrm{tr}\,U^4$; unweighted L^2 bounds alone would not suffice.

Since $r = O(p^{-1})$, $a_m = O(p)$, $c_m = O(1)$, and $m = O(p^2)$, every term on the right of (21) is $O_{L^2}(p^{-1})$ or smaller. In particular $a_mY/m = O_{L^2}(p^{-1})$; the potentially finite conserved contribution of A^2/m was already included in $D_{2,m}$ and has not been dropped. Projection is contractive and annihilates $D_{2,m}$, completing (3) for the raw coefficient.

Ordinary chaos is orthogonal only for the equal-time annealed inner product. Inserting e^{ith} or powers of h changes the Gaussian degree bookkeeping. Thus the Q_4 autocorrelation known from the earlier work does not by itself determine the full residual spectrum: the T_N autocorrelation and its energy-resolved cross with $Q_{4,N}$ must still be calculated.

7 Checks and the next spectral question

The symbolic record contains the exact five-word matrix, all rational filters, their full rank-proportion expansions, and the completed second-chaos table. Independent even polarization of the raw quartic definitions verifies 3,600 contractions in 720 directions. The first two systems, $(N, p, m) = (3, 1, 2), (4, 2, 3)$, cover every quadratic Gaussian coefficient; three additional directions use the larger half-filled system $(6, 2, 3)$. All entries are checked dyadic integers within the exact arithmetic range before comparison. A further 720 identities check the raw core decomposition.

Second contraction gives the five scalar means. Their independent rank sums agree in 10 cases. The $(4, 2, 3)$ sector has a lower-only rank absent from the upper sector, so it is outside the common-rank filter domain; the attempted mean check exposed this and was recorded. Its direct CAR contractions remain valid. The eventual half-filled limit lies inside the common-rank domain.

The assembly certificate also checks 123 exact charge-trace identities, the leading logarithm in (12), the fourth-chaos projection equations, the determinant (14), and the two small-λ limits. The trace-limit proof uses the balanced-singleton and absolute-tail argument, not these finite identity checks alone. An initial symbolic parsing failure involving the name `lambda` is retained alongside the corrected run.

The next task is the fixed-half-filled joint two-energy measure of T_N and $Q_{4,N}$. This will decide how the two positive equal-time pieces combine in the actual memory spectrum. A central-charge extension of the second-chaos filter and the finite-memory remainder are subsequent steps toward the bulk singlet/meson dictionary.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759, Sec. 2.

[2] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Secs. 2.1 and 3.1. The spectral Wick product is defined in Eq. (3.2).

[3] Alan Day and Chengheng, local project notes: *Second memory coefficient and quartic singlet*; *Quadratic contractions of the memory core*; *Two-charge memory filters*; *Fixed-charge residual norm*; *The quadratic memory core in the double-scaled limit* (6 November 2026). Source paths are listed in the accompanying literature record.